

IDR305 21H Introduction to Economics for Sport Management

Exam

Instructions (read carefully)

- 3 hours and 30 minutes, including all practical and technical actions needed to upload in Inspira.
- All supporting materials permitted. It is not allowed to cooperate with or receive help from others.
- Justify all your answers with economic concepts, logical reasoning, and love for sport.

Good Luck!

Question 1 (20 points)

Suppose there is a country that can produce two goods: X and Y. The country has 300 machines and 500 workers. Each machine can produce 4 units of X or 2 units of Y. Each worker can produce 1 unit of X and 3 units of Y.

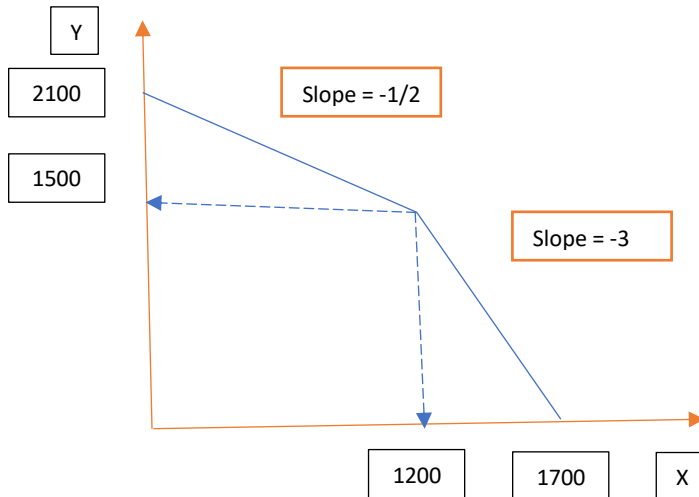
- Please draw a production possibilities frontier of this country and mark all the points (X and Y) where the curve changes its slope as well as the maximal production of X and Y. **(8 points)**
- Suppose that the country needs to consume 1500 units of Y. in this case, how many units of X will it consume? **(3 points)**
- Suppose there is a world trade, such that 1 unit of X is worth 4 units of Y. How many units of X and Y should the country produce and why? **(4 points)**
- Suppose that during the world trade, the country still needs to consume the number of units of X that you found in question b. in this case, how many units of Y will it consume? **(5 points)**

Answer

	Machines	Workers
Production X per one unit	4	1
Production Y per one unit	2	3
MC in producing X in terms of Y	0.5	3
MC in producing Y in terms of X	2	0.33
Max X	$300 \cdot 4 = \mathbf{1200}$	$500 \cdot 1 = 500$
Max Y	$300 \cdot 2 = 600$	$500 \cdot 3 = \mathbf{1500}$

- See graph below
- The country will consume 1200 units of X as shown in the graph below.
- The country has a comparative advantage in producing X, since its cost of producing X is at most 3Y. However, it can sell it to the world for the price of $X = 4Y$. Therefore, the country will only produce 1700 units of X and zero of Y.

- d. If the country needs to consume 1200 units of X, then it will produce 1700 units of X. The country will trade $1700 - 1200 = 500$ units of X for $500 \cdot 4 = 2000$ units of Y.



Question 2 (20 points).

You are given the following functions:

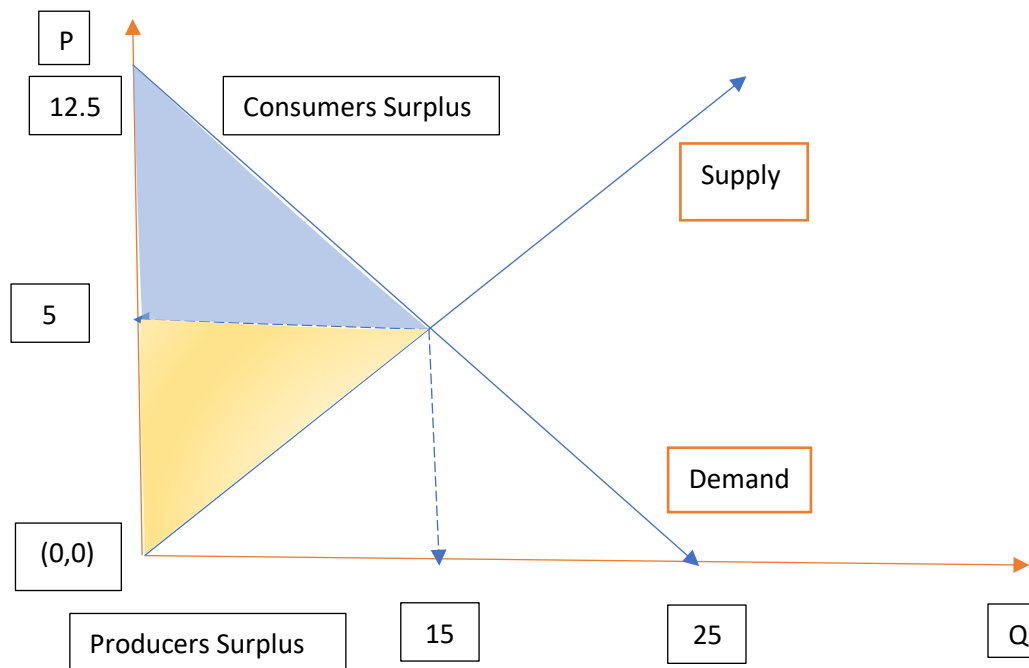
- 1) Demand function: $Q^D = 25 - 2p$.
- 2) Supply function: $Q^S = 3p$

Where p is the price of a good in US dollars (\$) and Q^D and Q^S are the quantity demanded and supplied, respectively for that specific good.

- a. What would be the price and the quantity in the equilibrium? **(4 points)**
- b. Show on the graph (where appropriate) the cases where $p=0$, $Q=0$, and price and quantity in equilibrium. **(8 points)**
- c. What is the producer surplus? **(3 points)**
- d. What is consumer surplus? **(3 points)**
- e. What is the total social welfare? **(2 points)**

Answer

- In equilibrium, supply equals demand $Q^D = Q^S$. Therefore, $25 - 2p = 3p$. $p = 5$ and $q = 15$
- See graph below
- The producer surplus is given by the area between the supply curve and the equilibrium price.
Producer Surplus (PS) : $\frac{[(5-0)*(15-0)]}{2} = \37.5
- The consumer surplus is given by the area between the demand curve and the equilibrium price.
Consumer Surplus (CS) : $\frac{[(12.5-5)*(15-0)]}{2} = \56.25
- In this case, social welfare is the sum of producer surplus and consumer surplus. $CS + PS = \$37.5 + \$56.25 = \$93.75$



Question 3 (20 points)

You are given the following firm's production function:

TP (Q)	L	MP	FC	VC	TC	MC	ATC	AVC
0	0				50			
1	10							
2	18							
3	30							
4	45							

One unit of labor (L) costs 20\$

- Complete the table. **(10 points)**
- What is the minimal price that the firm will produce in the short term? **(2 points)**
- What is the minimal price that the firm will produce in the long term? **(2 points)**
- It is known that each unit of production can be sold in the market for 240\$ ($p=240$). What would be the production of the firm? **(2 points)**
- What would be its profit in the short term? **(2 points)**
- What would be its profit in the long term? **(2 points)**

Answer

a.

TP (Q)	δQ	L	δL	MP	FC	VC	TC	MC	ATC	AVC
0	0	0	0	0	50	0	50	0	0	0
1	1	10	10	0.1	50	200	250	200	250	200
2	1	18	8	0.125	50	360	410	160	205	180
3	1	30	12	0.08	50	600	650	240	216.6	200
4	1	45	15	0.06	50	900	950	300	237.5	225

- Min AVC=180. Below that price the firm will not produce in the short term.
- Min ATC=205. Below that price the firm will not produce in the long term
- The rule of production says that $MC = P$. If $P=240$ \$, the firm will produce 3 units of Q
- Profits** (short-run) $= P \cdot Q - VC = 240 \cdot 3 - 600 = 120$ \$

In the short run, fixed costs are sunk costs, because they are in the past and cannot be changed. Therefore, they play no role in economic decisions about future production or pricing.

Since in the short run the profits are positive, the firm will produce.

f. Profits (long-run) = $P \cdot Q - TC = 240 \cdot 3 - 650 = 70\$$

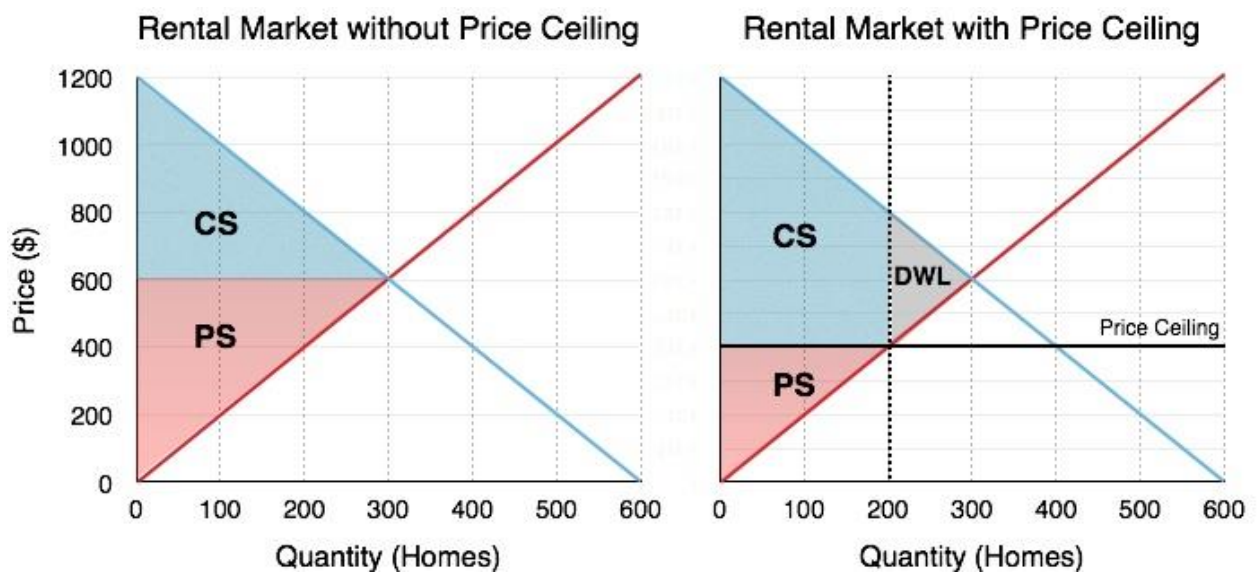
All costs are variables and therefore all the costs (fixed and variable) have to be taken into account

Since in the long run the profits are still positive, the firm will keep producing.

Question 4 (10 points)

- Show on a graph the effects of a price ceiling (5 points)
- Provide a sport-related policy that embodies the concept of price ceiling. (3 points)
- How are these two policies similar? How do they differ? (2 points)

Answer



a.

- Since salary cap is a rule that limits the amount of money a team can spend on players' wages, it can be seen as a sort of price ceiling.
- Salary cap and price ceiling, however, serve two different purposes: while price ceiling keeps a price from rising, salary cap tends to limit income disparity and promote competitive balance.

Question 5 (10 points)

There is a market with 7 workers who should be allocated between two fields, given the following production function:

Field A

L	TP
0	0
1	7
2	13
3	18
4	22
5	23

Field B

L	TP
0	0
1	4
2	7
3	9
4	10
5	10

- How many workers would work in each field if it is known that the price of one produced unit in field **A** is **1\$** and the price of one produced unit in field B is **3\$**? **(6 points)**
- What would be the wage in this market? **(4 points)**

Answer

Field A

L	TP	MP (Marginal Product)	Price (\$)	VMP (Value Marginal Product)	ΣVMP
0	0	0	1		
1	7	7	1	7	7
2	13	6	1	6	13
3	18	5	1	5	18
4	22	4	1	4	22
5	23	1	1	1	23

Field B

L	TP	MP (Marginal Product)	Price (\$)	VMP (Value Marginal Product)	ΣVMP
0	0		3		
1	4	4	3	12	12
2	7	3	3	9	21
3	9	2	3	6	27
4	10	1	3	3	30
5	10	0	3	0	30

- The optimal combination is to have 4 workers on field A and 3 workers on field B.
- The wage is the value of the marginal product of the last worker, which is 4.

Question 6 (5 points)

You are expected to receive a one-time sum of 100 NOK at the end of a five-years investment plan. What is the Present Value (PV) of this plan if the annual interest rate is 3% in the first two years and 5% per year after the first two years?

Answer

$$PV = \frac{100}{(1+0.03)^2 \cdot (1+0.05)^3} = 81.42$$

Question 7 (5 points)

You are expected to invest a one-time sum of 1000 NOK at the end of the sixth year from now and receive it at the end of the eighth year from now. What is the Future Value of this sum at the end of the eighth year from now if the annual interest rate is 3% in the first two years, 4% in the third year, 7% in the fourth year and 10% per year after that?

Answer

$$FV = 1000 * (1 + 0.1)^2 = 1210$$

Using Excel's formula: =FV(0.1,2,-1000)

Question 8 (5 points)

Suppose that tickets for the final of the FIFA World Cup 2022 in QATAR are a substitute good for an organized 3-days trip in Tromsø to see the Northern Light. What would happen to demand for this 3-days trip if the price for the tickets to the final of the World Cup increases?

Answer

An increase of the price for the ticket increase the demand for the 3-day trip.

Question 9 (5 points)

Define the concept of "competitive balance" in sports economics and mention one possible way to promote it.

Answer

Competitive balance assesses the condition of having an equal and fair competition. League commissioners endeavor to increase competitive balance in their league to keep players and fans engaged. One way to promote competitive balance is the use of salary cap.